

Evolutionary Computation and its Applications

Today's goals



Module 1

Fundamental Concepts in Optimization

Module 2

Genetic Algorithm

Module 5 Introduction

✔

Introduction to Module 5's Study

Video • 1 min

✔

How to Access MATLAB Online

Reading • 5 min

Module 5-1 Genetic Algorithm 1

✔

Genetic Algorithm: A First Introduction

Video • 6 min

✔

Genetic Algorithm: A Demo

Video • 8 min

✔

Materials on Genetic Algorithms

Reading • 10 min

✔

How Genetic Algorithms Solve Optimization Problems

Reading • 5 min

Module 5-2 Genetic Algorithm 2

✔

Genetic Algorithms: GA Representation, Initialization and Selection

Video • 10 min

✔

Genetic Algorithms: GA Crossover and Mutation

Video • 7 min

✔

Genetic Algorithms: GA Stopping Criterion and Summary

Video • 5 min

✔

Materials on Genetic Algorithms

Reading • 10 min

✔

How GA Operators Work Together

Reading • 5 min

Module 5-3 Use Genetic Algorithm 1

✔

Use GA via MATLAB: GA Function

Video • 7 min

✔

Use GA via MATLAB: Employ the Function

Video • 6 min

✔

Materials on Using Genetic Algorithms

Reading • 10 min

●

Using Genetic Algorithms Effectively in MATLAB

Reading • 5 min

Module 5-4 Use Genetic Algorithm 2

✔

Use GA via MATLAB: Case Studies

Video • 5 min

✔

Materials on Case Studies

Reading • 10 min

✔

Solving Real Engineering Problems with Genetic Algorithms

Reading • 5 min

Module 5 Quiz

✔

Module 5 Quiz

Practice Assignment • Grade: 100%

Module 5 Assignment

●

Module 5 Assignment

Graded App Item • 30 min

●

Objective function for Assignment 5

Reading • 5 min

✔

Applying Genetic Algorithms to a Real Engineering Design Problem

Reading • 5 min

●

Designing and Applying a Genetic Algorithm to Solve an Engineering Optimization Problem

Dialogue • 5 min

●

Module 5 Recap

Reading • 5 min

Module 3

Particle Swarm Optimization

Applying Genetic Algorithms to a Real Engineering Design Problem



0:03 / 0:52



2x





In this assignment, you used a genetic algorithm to optimize a microwave filter coupling matrix so that multiple frequency-domain performance requirements are satisfied.

Reflect on the following questions:

- How were the **design requirements** translated into constraints, and why was the **sum of constraint violations** chosen as the objective function?
- Why is achieving a **fitness value of zero** a meaningful success criterion in this optimization problem?
- How did the choice of **bounds, population size, and maximum generations** balance solution quality with computational cost?
- Based on the initial and optimized frequency responses, what insights can you draw about how small changes in coupling coefficients affect filter performance?

This discussion will help you connect genetic algorithm mechanics with real engineering decision-making and validation.

Go to next item

✔ Completed

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https://www.coursera.org/learn/evolutionary-computation-and-its-engineering-applications/supplement/dQM9L/applying-genetic-algorithms-to-a-real-engineering-design-problem

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